

Tushar Kanta Sahoo

DOCTORAL STUDENT

Department of Physics, Indian Institute of Science Bangalore - 560012, India

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“More Is Different” ... Philip W. Anderson

Education

Indian Institute of Science Bangalore

Karnataka, India

PHD IN PHYSICS

2025 -

- Working in Ultrafast Lightwave Quantum Electronic Group

Indian Institute of Technology (ISM) Dhanbad

Jharkhand, India

M.SC. IN PHYSICS

2023 - 2025

- CGPA - 9.3 [Transcript]

Utkal University

Odisha, India

B.SC. IN PHYSICS

2020 - 2023

- CGPA - 8.42 [Transcript]

Sri Aurobindo Science College

Odisha, India

HIGHER SECONDARY EDUCATION

2018 - 2020

- Percentage - 85.0 [Transcript]

Research Experience

M.Sc. Thesis

IIT (ISM) Dhanbad

RESEARCH ON THE PYGMY DIPOLE RESONANCE (PDR) IN EXOTIC NUCLEI

2024 - 2025

- Investigated the nature of PDR in nuclei to classify it as a single particle-hole (p-h) transformation or a collective excitation.
- Conducted simulations of transfer reactions to identify PDR states in ^{90}Zr .
- Utilized advanced nuclear simulation tools, including CHUCK3, AngCor, and ROOT, to analyze and interpret data on PDR states.
- Aimed to contribute to the understanding of nucleosynthesis processes in heavy, neutron-rich stars by utilizing PDR findings.

Achievements

GATE (Graduate Aptitude Test in Engineering)

2025

ALL INDIA RANK - 223 [SCORECARD]

CSIR NET (Council of Scientific and Industrial Research National Eligibility Test)

Dec, 2024

ALL INDIA RANK - 223 [AWARDED JUNIOR RESEARCH FELLOWSHIP (JRF)] [SCORECARD]

Bhabha Atomic Research Centre OCES/DGFS

2025

EXAM QUALIFIED AND SELECTED FOR INTERVIEW [SCORECARD]

JAM (Joint Admission Test for Masters)

2023

ALL INDIA RANK - 1029 [SCORECARD]

Academic Training

Workshop on Cavity Quantum Materials

IISc, Bangalore

PARTICIPANT [CERTIFICATE]

Mar, 2026

Winter School

IISc, Bangalore

PARTICIPANT [CERTIFICATE]

Dec, 2025

Technical Skills

Programming Languages

PROFICIENT IN A VARIETY OF PROGRAMMING LANGUAGES INCLUDING

- Python (Intermediate)
- Matlab (Intermediate)
- C++ (Basic)
- CHUCK3 (Basic)
- AngCor (Basic)
- LaTeX (Intermediate)

Design and Visualization Tools

EXPERIENCED IN USING DESIGN AND VISUALIZATION TOOLS SUCH AS

- SolidWorks
- Inkscape

Spoken Languages

COMFORTABLE TO SPEAK IN

- English, Hindi, Odia

Relevant Coursework In B.Sc. and M.Sc.

Quantum Mechanics, Electro Dynamics, Classical Mechanics, Mathematical Physics, Nuclear and Particle Physics, Statistical Mechanics, Atomic and Laser Physics, Advanced Condensed Matter Physics, Advanced Quantum Mechanics, Computational Physics.

Skills

- Proficient in experimental physics techniques, data Analysis, and equipment operation.
- Deep understanding of both theoretical physics concepts and mathematical methods.
- Curious and flexible to learn new concepts, skill and techniques.